1. UWINEZA Lidia (25RP00738)
2. SHEMA KABIRIGI Denny (25RP00642)
3. NIYONSENGA Eric (25RP02023)
4. UWINEZA Rebecca (25RP00446)

Assignment1

1.Study of Passive Components

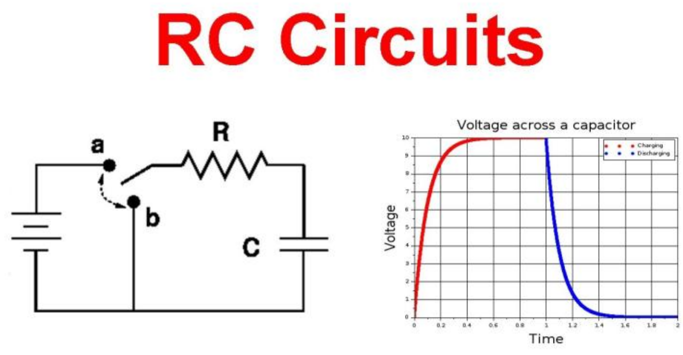
Feature	Resistor (R)	Capacitor (C)	Inductor (L)
Definition	A component that opposes the flow of electric current, converting electrical energy into heat.	A component that stores electrical energy in an electric field between two conductive plates.	A component that stores energy in a magnetic field when electric current flows through it.
Primary Property	Resistance (R): Opposition to current flow.	**Capacitance (C):** Ability to store charge.	**Inductance (L):** Opposition to changes in current.
Unit of Measurement	Ohm (Ω)	Farad (F)	Henry (H)
Behavior in DC	Opposes current according to Ohm's Law (V=IR).	Opposes current according to Ohm's Law (V=IR).	Acts as a **Short Circuit** (wire) in steady state.

Behavior in AC	Resistance is constant regardless of frequency.	Impedance decreases as frequency increases	mpedance increases as frequency increases
Coupling: Series	Req=R1+R2	Inverse Additive	Leq=L1+L2
Coupling: Parallel	Inverse Additive	Ceq=C1+C2	Inverse Additive
Applications	Current limiting, voltage division, biasing active devices, heating elements.	Filtering noise, decoupling, coupling AC signals, timing circuits, energy storage.	Transformers, chokes, motors, filters, tuned circuits (radios).
Ratings	Resistance value, Power rating (Watts), Tolerance	Capacitance, Working Voltage (DC/AC), Temperature coefficient	Inductance, Current rating (Amps), Q-factor.

2.circuits for each combination of the three passive devices

Problem1 :

A 10k-Ohm resistor and a 100uF capacitor are connected in series to a 12V DC source. Calculate the time constant (Tau) and the voltage across the capacitor after one time constant.



Solution:

* Formula: Tau = R * C *

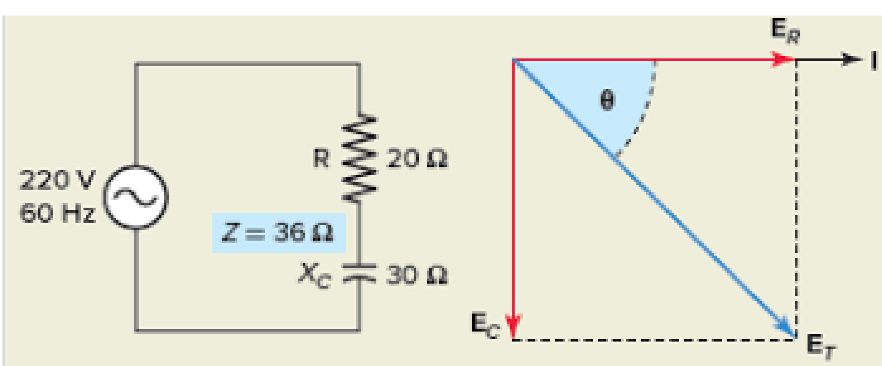
Tau = 10,000 Ohms * 0.0001 Farads = 1.0 second.

* Voltage Rule: After 1 Tau, a capacitor charges to 63.2% of the source voltage.

* Vc = 12V * 0.632 = 7.58V

Problem 2:

A 50-Ohm resistor and a 10uF capacitor are in series with a 50Hz AC source. Find the capacitive reactance (Xc) and total impedance (Z).



Solution:

Reactance Formula: Xc = 1 / (2 * pi * f * C)

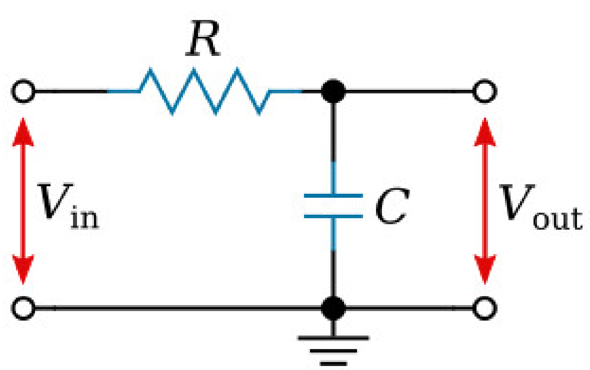
Xc = 1 / (2 * 3.1416 * 50 * 0.00001) = 318.3 Ohms

Impedance Formula: Z = sqrt(R^2 + Xc^2)

Z = sqrt(50^2 + 318.3^2) = 322.2 Ohms

Problem3:

ou need a low-pass filter with a cutoff frequency (fc) of 1kHz. If you use a 100nF capacitor, what resistor value is needed?



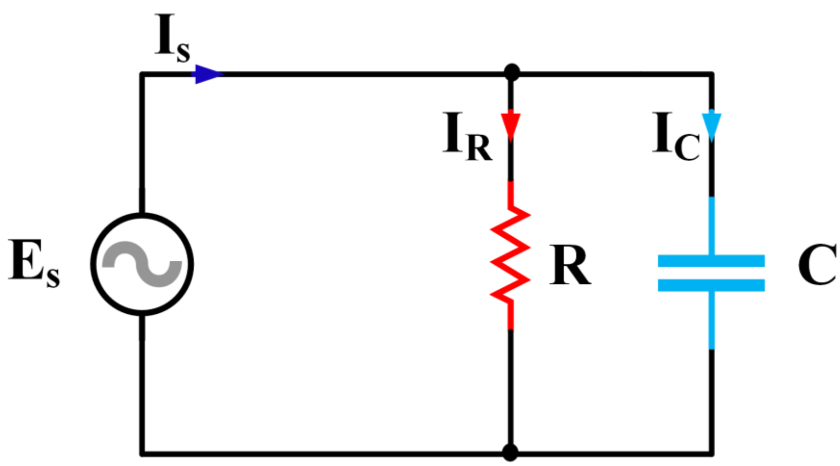
Solution:

Formula: fc = 1 / (2 * pi * R * C)

R = 1 / (2 * pi * fc * C) * R = 1 / (2 * 3.1416 * 1000 * 0.0000001) * R = 1591 Ohms (approx 1.6 kOhms)

Problem 4:

A 100-Ohm resistor and a capacitor with 100-Ohm reactance (Xc) are connected in parallel across 10V AC. Find the total current.



Solution:

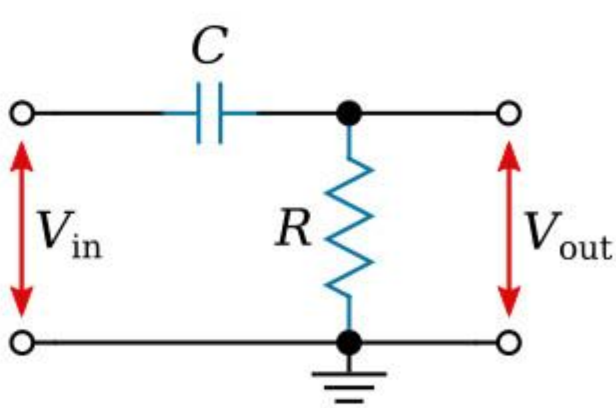
Current in Resistor (Ir): V / R = 10 / 100 = 0.1A

Current in Capacitor (Ic): V / Xc = 10 / 100 = 0.1A

Total Current Formula: Itotal = sqrt(Ir^2 + Ic^2) * Itotal = sqrt(0.1^2 + 0.1^2) = 0.141A

Problem 5:

A high-pass filter has R = 1k-Ohm and Xc = 1k-Ohm at a specific frequency. The input voltage is 10V. What is the output voltage?



Solution:

This acts as a voltage divider.

Formula: Vout = Vin * (R / Z)

Impedance (Z) = sqrt(1000^2 + 1000^2) = 1414.2 Ohms

Vout = 10 * (1000 / 1414.2) = 7.07V